specific problem. More and more data can be collected to make the model more efficient.

Testing Dataset

After training and validation of the model, a metric is needed to test the model. An unbiased measure to calculate the predictive power of the model is required. The model should have made the right generalizations. The optimum standard to evaluate a model is the testing dataset. It is a well curated dataset containing samples of all possible scenarios that can mimic the 'real world' the machine will actually run in. If validation dataset is not used to tune the model, then the validation results will reflect the test results.

Validation Dataset

A validation dataset is a sample of data held back from training your model that is used to give an estimate of the skill of the model while tuning the hyperparameters. To get an easier understanding, have a look at how an expert in the field defines it:

"Suppose that we would like to estimate the test error associated with fitting a particular statistical learning method on a set of observations. The validation set approach [...] is a very simple strategy for this task. It involves randomly dividing the available set of observations into two parts, a training set and a validation set or hold-out set. The model is fit on the training set, and the fitted model is used to predict the responses for the observations in the validation set. The resulting validation set error rate — typically assessed using MSE in the case of a quantitative response—provides an estimate of the test error rate."

— Gareth James, et al., Page 176, An Introduction to Statistical Learning: with Applications in R, 2013.

The validation data tests whether the solution is capable of being called an abstraction of learning.

Hyperparameters

We mentioned hyperparameters every time validation set was touched upon. The definition of hyperparameters is often abstract and flexible. Machine learning models need to be optimised in order to reduce the testing error percentage. Hyperparameters are critical as they can tune the behaviour of the machine learning model. They are hidden elements not directly tied to the model but which can heavily influence it. They are external to the model and their values cannot be predicted using data. In general,